

# Dhruv Maheshwari

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## EDUCATION

### PES University, Bengaluru

2027

Bachelor of Technology - **Computer Science and Engineering**

Prof. CNR Rao **Scholarship** (40% Tuition Merit Award), **4× Distinction Award**

## SKILLS

**Languages:** Python, SQL

**Frameworks & Libraries:** FastAPI, Uvicorn, FastEmbed, LangChain, LangGraph, ChromaDB, Three.js

**Tools & Platforms:** Git, GitHub, Docker, Kubernetes, AWS Bedrock, Agno, Vercel, Render, Azure Dev Console, HuggingFace, Groq, Blender

**Concepts:** API design, Multi-agent AI systems, Retrieval Augmented Generation (RAG), LLM inference, Embeddings, Knowledge Graphs, CI/CD, Prompt Engineering

**Coursework:** DSA, OS, DBMS, ML, Computer Networks

## EXPERIENCE

### Intern – Technology

Jun – Aug 2026

Acenet Consulting

*Python · AWS Bedrock · FastAPI · Uvicorn · Pandas · NumPy · Azure Deployment · LangChain*

- Designed and developed a **multi-agent** AI-powered Virtual CFO platform using Python, FastAPI, and AWS Bedrock to deliver macroeconomic, industry, and automotive market intelligence.
- Built **automated data ingestion, cleaning, and historical analytics pipelines**; integrated LLM-based reasoning agents for executive insights; implemented intelligent query routing across specialized domains.
- Developed **REST APIs**, scenario analysis, and **confidence scoring** with automated reporting, optimizing data processing workflows to support data-driven business decisions.

### AI App Development Intern

Jun – Jul 2025

xForm Solutions

*Python · PyTorch · NumPy · BERT*

- Built a **multilingual speech understanding system** (Whisper + NLLB-200 + SBERT) achieving **sub-0.07 Character Error Rate (CER)** on English benchmark speech with semantic consistency across Hindi/Tamil code-mixed inputs.
- Conducted **systematic WER/CER evaluation** on the **Google FLEURS dataset**, optimizing the preprocessing pipeline to improve **non-English transcription accuracy**.
- Engineered **rule-based urgency detection** and post-processing logic to improve semantic precision in **noisy, low-resource speech environments**.

## PROJECTS

### pRAGma - RAG Application

*Python · FastAPI · React (Vite) · Prompt Engineering · FastEmbed · Groq API*

[github.com/dhruvm-04/pRAGma](https://github.com/dhruvm-04/pRAGma)

- Built a **full-stack RAG application** around published IR/RAG papers, implementing **HyDE**, **RAG-Fusion**, and **Step-Back**.
- Exposed **retrieved chunks**, similarity scores, source snippets, latency, and token usage for **explainable retrieval**.
- Added pipeline visualization for **Direct**, **Multi Query**, **RAG-Fusion**, **Decomposition**, **Step-Back**, and **HyDE**.

aiCFO - Multi Agent Advisory Application

Python · LangChain · LangGraph · Groq · Flask · Pandas · yFinance [github.com/dhruvm-04/aiCFO](https://github.com/dhruvm-04/aiCFO)

- Engineered a **parallel multi-agent workflow** using **LangGraph**, routing a single query to specialized Automotive and Macroeconomic agents before synthesis by a CFO agent.
- Implemented **LLM-driven tool calling** across **16 domain-specific tools** spanning government automotive and economic datasets, enabling agents to dynamically select tools for trend and metric analysis.
- Built **schema-constrained agent communication** and strict JSON-schema outputs, producing typed analyses with findings, trends, data points, caveats, and recommendations.

MusiConvert - Secure Peer-to-Peer Playlist Sharing Platform

Python · Socket Programming · Fernet Encryption · Multithreading [github.com/dhruvm-04/MusiConvert](https://github.com/dhruvm-04/MusiConvert)

- Implemented a **custom Selective RepeatARQ protocol** from scratch over **raw UDP sockets** - per-packet **sequencing**, **sliding windows**, threaded **retransmission timers**, and cumulative **ACKs** - to reliably transfer playlists directly between peers on a local network.
- Designed a fuzzy-matching engine that scores candidate tracks using weighted title/artist string similarity plus a duration-tolerance check, so mismatched metadata across platforms **doesn't produce false matches**.

EchoLang - Real-Time Multilingual Speech Translator

Python · PyTorch · OpenAI Whisper · Sentence-BERT · Transformers [github.com/dhruvm-04/EchoLang](https://github.com/dhruvm-04/EchoLang)

- Developed an end-to-end **ASR → Machine Translation → Intent Classification pipeline** for code-switched Hindi, Tamil, and English speech using Whisper, NLLB-200, and Sentence-BERT.
- Systematically **evaluated the pipeline on the Google FLEURS dataset** to improve accessibility for blue-collar workers across India, achieving sub-0.07 CER (Character Error Rate) on English benchmark speech.

| CERTIFICATIONS                                                                   | Link            |
|----------------------------------------------------------------------------------|-----------------|
| • <b>Google</b> IT Automation with Python - OOPs · Python · DSA                  | <i>Jun 2023</i> |
| • <b>AWS</b> Prompt Engineering Foundation - Prompt Engineering · Responsible AI | <i>Mar 2025</i> |
| • <b>IBM</b> Fundamentals of AI - Machine Learning · Ethical AI                  | <i>Jan 2025</i> |

EXTRACURRICULARS

- Taught **3D modeling** and **game-ready asset creation** to ~50 students as **Blender Workshop Instructor**, PES University.
- **Led chassis design and fabrication** for the automotive club **Team Vegavath** using Blender and Fusion360.
- Served as **Under Secretary General** and led the Design team for PES University's **Model United Nations society MUNSoc PESUECC**.
- **Led the Design team** for **Maaya 2025**, PES University's **annual techno-cultural fest**.